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Incentive Constrained Risk Sharing, Segmentation, and Asset Pricing

Contents

1 Research statement

Markets can look "complete" in terms of the span of securities, yet risk sharing can remain incomplete because agents cannot pledge (or creditors cannot seize) the full payoff of the assets used as collateral. This imperfect pledgeability endogenously restricts how much state-contingent debt agents can issue. As a result, agents have *different effective state prices* (private valuation operators), so different agents optimally hold different assets.

That is: heterogeneous incentives $\Rightarrow$ heterogeneous marginal rates of substitution $\Rightarrow$ heterogeneous asset valuations $\Rightarrow$ **endogenous segmentation** & **basis**.

2 Model (Section I)

2.1 Time and uncertainty

- Finite horizon $t \in \{0, 1, \dots, T\}$.
- Each period, a state $s_t \in S$ is drawn from a finite set S.
- A history is $s^t = (s^{t-1}, s_t)$ and the set of time-t histories is S^t.
- $\pi_t(s^t) > 0$ denotes the probability of history s^t conditional on s_0.
- A node in the event tree is (t, s^t).

2.2 Assets: one-step Arrow securities and a (potentially large) set of trees

At each node (t, s^t) with $t < T$:

- A complete set of **one-step-ahead Arrow securities** is traded in **zero net supply**. Let $Q_{t+1}(s^t, s)$ denote the price at (t, s^t) of the Arrow security paying 1 unit at $(t + 1, (s^t, s))$.
- In addition, there are **trees** in positive supply. A tree is a dividend stream

$$\delta \equiv \{\delta_t(s^t) : t \geq 0,\ s^t \in S^t\}, \qquad \delta_t(s^t) \in [0, \bar{\delta}].$$

  The set of all dividend streams is Δ (no special structure beyond boundedness).

- The supply of trees is represented by a **finite positive measure** $\bar{N}$ on Δ. This unifies cases with finitely many trees (discrete support) and rich asset menus (continuous support).

Aggregate dividend positivity assumption. The paper assumes that aggregate dividends are strictly positive at every node:

$$\int_\Delta \delta_t(s^t)\, d\bar{N}(\delta) \;>\; 0 \qquad \text{for all } (t, s^t). \tag{1}$$

This technical condition is used for equilibrium existence and to avoid degenerate "zero aggregate consumption" nodes.

2.3 Agents and preferences

There is a finite set of agent types indexed by $i \in \{1, \dots, I\}$, with unit measure of each type. Agent i chooses consumption plan $c_i = \{c_{it}(s^t)\}$ to maximize expected discounted utility:

$$\sum_{(t,s^t)} \beta^t \pi_t(s^t)\, u_i\big(c_{it}(s^t)\big), \tag{2}$$

where u_i is increasing, concave, continuous, and C^1 on $c > 0$ (allowing $u_i(0) = -\infty$ such as log utility).

2.4 Endowments

- At $t = 0$, agent i holds a fraction $\alpha_i > 0$ of the market portfolio of trees:

$$N_{i,-1} \equiv \alpha_i \bar{N}, \qquad \sum_{i=1}^{I} \alpha_i = 1.$$

- At each node (t, s^t) the agent receives a nonnegative endowment $e_{it}(s^t) \geq 0$ ("labor income").

2.5 Trading, budget constraint, and incentive constraint

2.5.1 Sequential budget constraint with Arrow securities

At node (t, s^t) $(t < T)$, agent i: consumes $c_{it}(s^t)$, buys a nonnegative tree portfolio (a positive measure) $N_{it}(s^t)$ over Δ, and takes Arrow positions $a_{it+1}(s^t, s)$ for each $s \in S$.

Let $P_t(\delta \mid s^t)$ be the tree price function at node (t, s^t). Then, for $t < T$, the sequential budget constraint is:

$$\begin{aligned} c_{it}(s^t) + \int_\Delta P_t(\delta \mid s^t)\, dN_{it}(\delta \mid s^t) + \sum_{s\in S} Q_{t+1}(s^t, s)\, a_{it+1}(s^t, s) \\ = e_{it}(s^t) + \int_\Delta \big[\delta_t(s^t) + P_t(\delta \mid s^t)\big]\, dN_{i,t-1}(\delta \mid s^{t-1}) + a_{it}(s^t). \end{aligned} \tag{3}$$

At terminal date T, the constraint becomes:

$$c_{iT}(s^T) = e_{iT}(s^T) + \int_\Delta \delta_T(s^T)\, dN_{i,T-1}(\delta \mid s^{T-1}) + a_{iT}(s^T). \tag{4}$$

2.5.2 Imperfect pledgeability and the incentive constraint

From $t \geq 1$, the agent can threaten to default. Creditors can only seize a fraction $(1-\theta)$ of the agent's *long* positions, where $\theta \in (0,1)$. Under two-sided limited commitment (renegotiation via take-it-or-leave-it), the agent's **state-contingent liabilities** cannot exceed a fraction $(1-\theta)$ of her long-position payoff.

Let $a^- = \max\{-a, 0\}$ and $a^+ = \max\{a, 0\}$. The constraint can be written (paper's starting point):

$$a^-_{it+1}(s^t, s) \leq (1-\theta)\Big\{a^+_{it+1}(s^t, s) + \int_\Delta \big[\delta_{t+1}(s^t, s) + P_{t+1}(\delta \mid s^t, s)\big]\, dN_{it}(\delta \mid s^t)\Big\}. \tag{5}$$

Since the constraint binds only for short positions, it simplifies to:

$$-a_{it+1}(s^t, s) \leq (1-\theta) \int_\Delta \big[\delta_{t+1}(s^t, s) + P_{t+1}(\delta \mid s^t, s)\big]\, dN_{it}(\delta \mid s^t). \tag{6}$$

Interpretation: liabilities must be backed by *pledgeable* collateral value; only $(1-\theta)$ of asset payoffs can be pledged.

2.5.3 Cash-on-hand reparametrization

Define cash-on-hand:

$$W_{it}(s^t) \equiv e_{it}(s^t) + \int_\Delta \big[\delta_t(s^t) + P_t(\delta \mid s^t)\big]\, dN_{i,t-1}(\delta \mid s^{t-1}) + a_{it}(s^t). \tag{1}$$

Then the budget constraint becomes:

$$\begin{aligned} &c_{it}(s^t) + \int_\Delta P_t(\delta \mid s^t)\, dN_{it}(\delta \mid s^t) + \sum_{s \in S} Q_{t+1}(s^t, s)\, W_{it+1}(s^t, s) \\ &\quad = W_{it}(s^t) + \sum_{s \in S} Q_{t+1}(s^t, s)\, e_{i,t+1}(s^t, s) + \sum_{s \in S} Q_{t+1}(s^t, s) \int_\Delta \big[\delta_{t+1}(s^t, s) + P_{t+1}(\delta \mid s^t, s)\big]\, dN_{it}(\delta \mid s^t). \end{aligned} \tag{7}$$

The incentive constraint becomes:

$$W_{it+1}(s^t, s) \geq e_{i,t+1}(s^t, s) + \theta \int_\Delta \big[\delta_{t+1}(s^t, s) + P_{t+1}(\delta \mid s^t, s)\big]\, dN_{it}(\delta \mid s^t). \tag{8}$$

Interpretation: next-period cash-on-hand must cover the *non-pledgeable* part of income (labor plus θ fraction of tree payoff).

2.6 Equilibrium definition

A price system is (P, Q), where $P = \{P_t(\delta \mid s^t)\}$ is a sequence of positive continuous functions on Δ and $Q = \{Q_{t+1}(s^t, s)\}$ is a sequence of Arrow prices.

A plan (c_i, N_i) is **budget feasible and incentive compatible** if there exists W_i such that (7) and (8) hold at all nodes and the initial condition holds:

$$W_{i0}(s_0) = e_{i0}(s_0) + \alpha_i \int_{\Delta} \big[\delta_0(s_0) + P_0(\delta \mid s_0)\big]\, d\bar{N}(\delta). \tag{9}$$

Feasibility / market clearing. At each node:

$$\sum_i c_{it}(s^t) = \sum_i e_{it}(s^t) + \sum_i \int_{\Delta} \delta_t(s^t)\, dN_{i,t-1}(\delta \mid s^{t-1}), \tag{10}$$

$$\sum_i N_{it}(s^t) = \bar{N}. \tag{11}$$

Competitive equilibrium. An equilibrium is a price system (P, Q) and feasible allocation $\{(c_i, N_i)\}_{i=1}^{I}$ such that each (c_i, N_i) solves agent i's optimization problem given (P, Q).

3 Equilibrium analysis (Section II)

3.1 No-arbitrage relationships (Lemma 1)

Even though standard frictionless no-arbitrage fails (because pledgeability matters), the paper derives three inequalities that must hold in any equilibrium:

Lemma 1 (Lemma 1: one-directional deviations from the law of one price)**.** *In equilibrium:*

(i) Arrow prices are strictly positive:

$$Q_{t+1}(s^t, s) > 0. \tag{12}$$

(ii) For trees in positive supply*, the tree price is* at most *the value of its total future payoff (replicating Arrow portfolio):*

$$P_t(\delta \mid s^t) \leq \sum_{s \in S} Q_{t+1}(s^t, s)\Big[\delta_{t+1}(s^t, s) + P_{t+1}(\delta \mid s^t, s)\Big], \qquad \bar{N}\text{-a.e. } \delta. \tag{13}$$

(iii) For all trees, the tree price is at least *the value of its* pledgeable *payoff:*

$$P_t(\delta \mid s^t) \geq (1-\theta) \sum_{s \in S} Q_{t+1}(s^t, s) \Big[\delta_{t+1}(s^t, s) + P_{t+1}(\delta \mid s^t, s) \Big], \tag{14}$$

with strict inequality when the continuation dividend is nonzero.

Intuition / proof sketch. (i) If some Arrow price were zero, agents would demand unbounded consumption at the corresponding node.
(ii) If a *supplied* tree were priced above its Arrow-replicating payoff, any holder could sell it, buy the Arrow portfolio, and increase current consumption *without tightening* incentive constraints (because substituting a tree by Arrow claims does not increase non-pledgeable income). Market clearing would fail.
(iii) If a tree were priced below the PV of its pledgeable payoff, an agent could buy it by issuing liabilities backed by the pledgeable part and keep strictly positive non-pledgeable payoffs as a free lunch, again violating market clearing.

Together, (13)–(14) imply **one-directional law-of-one-price violations**: trees can be cheap relative to Arrow replication (a "negative basis"), but cannot be expensive.

3.2 Existence (Theorem 1) and short-selling irrelevance (Corollary 1)

Theorem 1 (Theorem 1)**.** *An equilibrium exists.*

Proof architecture. The paper adapts the Arrow–Debreu price-player existence proof to: (i) allow incentive constraints that depend on prices and (ii) allow deviations from the law of one price in the tree market. The proof proceeds by first establishing existence for *finite-support* tree supplies (finite-dimensional price vectors), extending to continuous price functions, and then using weak convergence of measures to handle general $\bar{N}$.

Corollary 1 (Corollary 1)**.** *Any equilibrium with only Arrow short selling remains an equilibrium when agents can short both trees and Arrow securities.*

Intuition. Because (13) implies trees are priced *below* Arrow-replicating portfolios, agents do not want to short trees: shorting the Arrow-replicating payoff dominates (it is at least as cheap and is "more pledgeable" in the incentive constraint). Hence the tree short-selling constraint does not bind.

3.3 When do incentive constraints bind? (Proposition 1: IC-implementability)

Let (Q, c) denote a frictionless (complete markets, no incentive constraints) equilibrium. The paper defines:

> (Q, c) is **IC-implementable** if there exists an equilibrium with incentive constraints whose Arrow prices equal Q and whose consumption allocation equals c.

Proposition 1 (Proposition 1: conditions for IC-implementability)**.** *Let $\theta^\star$ be the largest pledgeability wedge θ for which a given frictionless equilibrium is IC-implementable. Then (qualitatively):*

(i) *$\theta^\star > 0$ if labor income is small and utilities satisfy Inada conditions (agents dislike low consumption so they avoid large liabilities).*

(ii) *$\theta^\star < 1$ if collateral supply $\bar{N}$ is scarce, labor income is sizable, and marginal rates of substitution evaluated at e are not equalized.*

(iii) *$\theta^\star < 1$ with heterogeneous CRRA, small labor income, and a* single *tree (too much bundling of aggregate risk makes full insurance infeasible under limited pledgeability).*

(iv) *$\theta^\star$ increases (weakly) when trees are* unbundled *into replicating portfolios (more collateral & hedging efficiency).*

Core intuition. Implementability is easiest when agents do not need to issue large state-contingent liabilities to replicate the frictionless consumption allocation. Scarce or heavily bundled collateral forces agents to rely on debt issuance for state-contingent transfers, which runs into the incentive constraint.

4 First-order conditions and private valuation operators (Section II.C)

4.1 Lagrangian multipliers

Let $\lambda_{it}(s^t) \geq 0$ be the multiplier on the budget constraint (7) at (t, s^t) and $\mu_{i,t+1}(s^t, s) \geq 0$ the multiplier on the incentive constraint (8) at successor node $(t+1, (s^t, s))$.

Assuming interior consumption for clarity, the paper obtains:

$$\beta^t \pi_t(s^t)\, u_i'\big(c_{it}(s^t)\big) = \lambda_{it}(s^t), \tag{16}$$

$$\lambda_{i,t+1}(s^t, s) + \mu_{i,t+1}(s^t, s) = \lambda_{it}(s^t)\, Q_{t+1}(s^t, s). \tag{17}$$

4.2 Arrow prices as wedges between IMRS and market state prices

Combining (16) and (17), the Arrow price can be written as:

$$Q_{t+1}(s^t, s) = \beta\, \pi_{t+1}(s \mid s^t)\, \frac{u_i'\big(c_{i,t+1}(s^t, s)\big)}{u_i'\big(c_{it}(s^t)\big)} + \frac{\mu_{i,t+1}(s^t, s)}{\lambda_{it}(s^t)}. \tag{18}$$

Interpretation: for unconstrained agents ($\mu = 0$), Arrow prices equal the agent's intertemporal marginal rate of substitution (IMRS). For constrained agents, the incentive constraint creates a positive wedge μ/λ.

4.3 Tree holdings and *private* Arrow valuations

The first-order condition for tree holdings implies that an agent evaluates the tree payoff using a *private valuation operator* that mixes market Arrow prices (pledgeable part) with the agent's IMRS (non-pledgeable part).

Specifically, define agent i's private Arrow valuation:

$$Q^i_{t+1}(s^t, s) \equiv (1-\theta)\, Q_{t+1}(s^t, s) + \theta\, \beta\, \pi_{t+1}(s \mid s^t)\, \frac{u_i'\big(c_{i,t+1}(s^t, s)\big)}{u_i'\big(c_{it}(s^t)\big)}. \tag{21}$$

Then the tree-holding condition can be written as:

$$\sum_{s \in S} Q^i_{t+1}(s^t, s)\Big[\delta_{t+1}(s^t, s) + P_{t+1}(\delta \mid s^t, s)\Big] \le P_t(\delta \mid s^t), \tag{20}$$

with equality whenever agent i holds the tree in positive quantity.

Key economic message. Imperfect pledgeability splits payoffs into:

- a **pledgeable component** (fraction $1-\theta$), valued at market state prices Q,
- a **non-pledgeable component** (fraction θ), valued at the agent's own IMRS.

Hence, even with identical beliefs, different binding patterns of incentive constraints create heterogeneous Q^i across agents.

5 Segmentation via optimal payoff sets (Section II.D)

5.1 From private valuations to an assignment problem

Fix a node (t, s^t) and consider a vector of one-period-ahead payoffs $x \in \mathbb{R}_+^{|S|}$. Agent i's private valuation is the dot product:

$$Q_{t+1}^i(s^t) \cdot x \equiv \sum_{s \in S} Q_{t+1}^i(s^t, s)\, x(s). \tag{22}$$

Define the **optimal payoff set** for agent i:

$$X_t^i(s^t) \equiv \Big\{ x \in \mathbb{R}_+^{|S|} :\ Q_{t+1}^i(s^t) \cdot x \geq Q_{t+1}^j(s^t) \cdot x \text{ for all } j \Big\}. \tag{23}$$

Agent i holds trees whose one-period-ahead payoff vectors lie in $X_t^i(s^t)$. Thus, determining who holds which assets is an **assignment** problem: allocate payoff vectors x to the agents who value them most (given Q^i).

5.2 Geometry (Proposition 2) and interpretation

Proposition 2 (Proposition 2: properties of optimal payoff sets)**.** *At each node (t, s^t):*

(i) Each $X_t^i(s^t)$ is a (possibly empty) ***convex polyhedral cone*** *in $\mathbb{R}_+^{|S|}$, and the cones form a partition (up to boundaries) of $\mathbb{R}_+^{|S|}$.*

(ii) If two agents have identical private valuations $Q_{t+1}^i = Q_{t+1}^j$ then $X^i = X^j$. If $Q^i \neq Q^j$, then the intersection of interiors is empty: $\mathring{X}^i \cap \mathring{X}^j = \emptyset$.

(iii) If no incentive constraint binds next period, all Q^i coincide and $X^i = \mathbb{R}_+^{|S|}$ (no segmentation). If some incentive constraint binds for some agent in some successor state, then some X^i are strict subsets and there exist $i \neq j$ such that $\mathring{X}^i \cap \mathring{X}^j = \emptyset$ (segmentation).

Why cones? The defining inequalities in (23) are homogeneous and linear in x, so scaling a payoff vector by a positive scalar does not change who is the best holder. Hence optimal sets are cones: they classify *risk exposures* (directions in payoff space), not payoff scale.

Connection to optimal transport / power diagrams. The collection $\{X^i\}$ corresponds to a *power diagram* in payoff space. Computationally, given the vectors Q^i, one can construct the cells X^i by solving a classical optimal transport / assignment problem: allocate each payoff vector to the agent with highest dot product valuation.

6 Parametric illustrations and interpretation of Figures 1–4

6.1 Figure 1: the set of dividend streams Δ in a simple event tree

What the figure shows. With two dates $t \in \{0, 1\}$ and two states $S = \{1, 2\}$, a dividend stream δ can be parameterized by $(\delta_0, \delta_1(1), \delta_1(2))$. Boundedness $\delta_t(\cdot) \in [0, \bar{\delta}]$ makes Δ a rectangular box in $\mathbb{R}^3$.

Why it matters. The paper's use of a measure $\bar{N}$ on Δ allows for a very rich asset menu—potentially including Arrow-like short-lived trees, long-lived trees, bonds of different maturities, etc. Segmentation is not driven by exogenous market incompleteness; it is driven by pledgeability constraints.

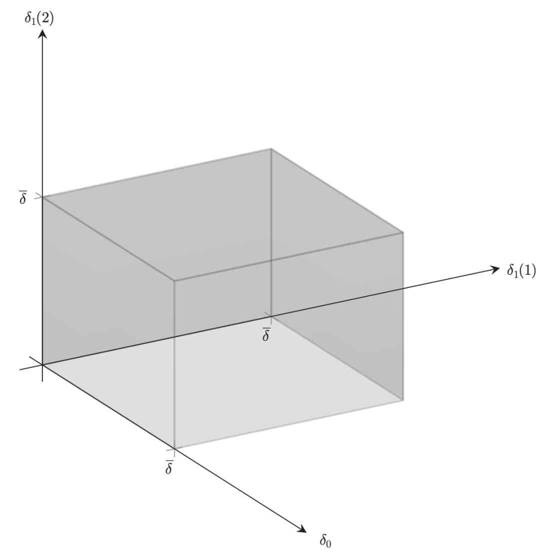

Figure 1. The Set Δ When There Are Two Periods, $t \in \{0,1\}$, and Two States, $S = \{1,2\}$

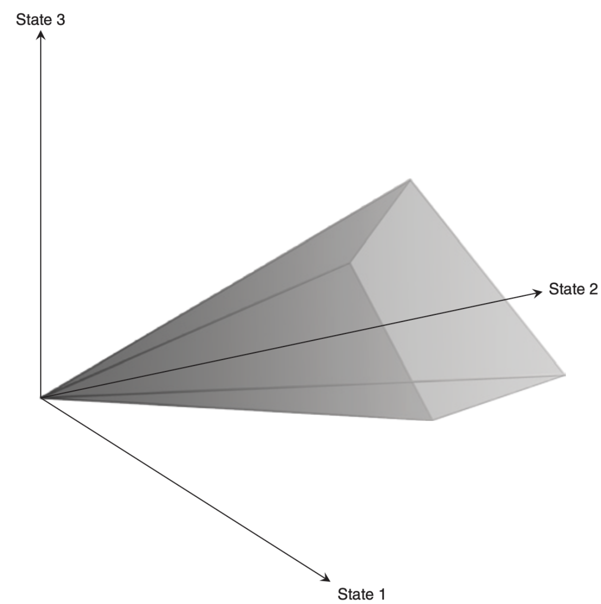

Figure 2. A Polyhedral Cone with Three Successor States

6.2 Figure 2: a polyhedral cone in payoff space

Object. At a node with three successor states, one-period-ahead payoffs live in $\mathbb{R}^3_+$. Proposition 2 implies each agent's optimal payoff set is a convex polyhedral cone.

Interpretation. The cone represents the set of payoff directions for which an agent is the best holder given her private state-price vector Q^i. Intersections of these cones with the simplex (payoffs normalized to sum to 1) provide a convenient 2D representation.

6.3 Figure 3: first parametric example (three states, log utility, no collateral limit)

The paper considers many log utility agents hit by heterogeneous i.i.d. endowment growth shocks. In the limiting economy with no collateral ($\bar{N} = 0$), the allocation is autarkic, so $c_{it}(s^t) = e_{it}(s^t)$. This makes each agent's IMRS and hence Q^i easy to compute, and the payoff-space partition can be plotted.

Reading the simplex partition. Each region on the simplex corresponds to one agent (type) being the best holder for payoffs with that state composition. Regions near the simplex corners correspond to Arrow-like payoffs (paying mostly in one state). Agents located near a corner are those with particularly high marginal utility (high state price) in that state. Other agents have regions away from corners: they are best holders of more diversified payoffs.

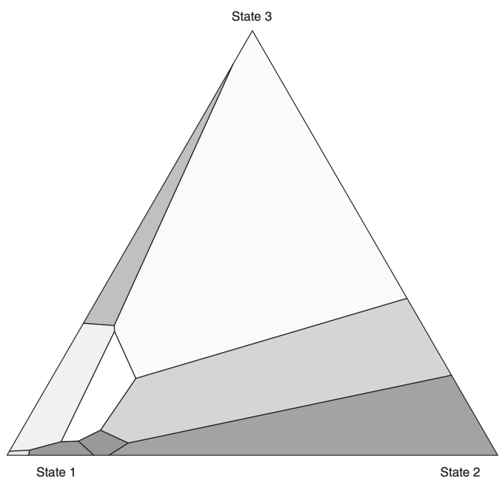

Figure 3. First Parametric Example: Intersections of Optimal Payoff Sets with the Simplex

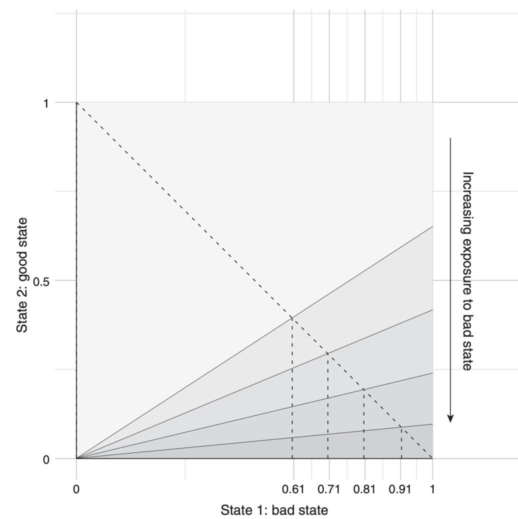

Figure 4. Second Parametric Example: Intersection of Optimal Payoff Sets with the Unit Square

6.4 Figure 4 and Proposition 3: a two-state economy with ordered types

In the second parametric example there are two states (bad $s = 1$ and good $s = 2$) and log utility agents indexed by exposure α. Endowment growth takes the form:

$$g_1(\alpha) = g(1 + k_1\alpha)^{-\phi}, \tag{24}$$

$$g_2(\alpha) = g(1 - k_2\alpha)^{-\phi}, \tag{25}$$

with parameters ensuring state 1 is "bad" and higher α increases bad-state exposure.

Proposition 3 (Proposition 3: ordered interval structure)**.** *Under the parameter condition stated in the paper, there exist cutoffs* $0 = x_0 < x_1 < \cdots < x_I = 1$ *such that the intersection* *of agent i's optimal payoff set with the simplex is the interval* $X_i = [x_{i-1}, x_i]$ *(where* x *is the payoff share in the bad state). Agents with higher bad-state exposure (higher* α*) hold trees with larger payoffs in the bad state, and extreme types hold assets close to Arrow payoffs.*

Interpretation of Figure 4. The horizontal axis is the bad-state payoff share $x \in [0, 1]$. Different agent types occupy adjacent intervals in x. As exposure to the bad state increases, the preferred/held interval shifts toward higher x (more insurance in the bad state). Thus, segmentation has an ordered "clientele" structure in this example.

7 Asset pricing implications (Section II.E)

7.1 Equilibrium pricing as an upper envelope of private valuations

Combining the tree-holding inequality (20) across agents yields the key pricing recursion:

$$P_t(\delta \mid s^t) = \max_i \sum_{s \in S} Q^i_{t+1}(s^t, s) \Big[\delta_{t+1}(s^t, s) + P_{t+1}(\delta \mid s^t, s)\Big]. \tag{26}$$

So the tree price is the **maximum** of agents' private valuations.

7.2 A one-directional basis (Proposition 4)

Because (26) is convex and homogeneous in payoffs, it implies subadditivity: bundling risks into one security can lower its price relative to an unbundled replicating portfolio. Formally:

Proposition 4 (Proposition 4: basis relative to replicating portfolios)**.** *Let* M *be a replicating portfolio of* long *positions (trees and/or Arrow securities) that matches the tree's one-period-ahead payoff. If no single agent is the best holder for (almost) all components of the replicating portfolio, then the tree is strictly cheaper than the replicating portfolio:*

$$P_t(\delta \mid s^t) < \int_{\Delta} P_t(\delta' \mid s^t)\, dM(\delta'). \tag{27}$$

Economic intuition. A single tree bundles payoffs across states in fixed proportions. A replicating portfolio allows those state payoffs to be unbundled and redistributed across different clienteles. If different agents value different states most, no one wants to hold the full bundle. Hence the bundled security is discounted relative to the unbundled replication.

Why the basis has a sign. Lemma 1 rules out the opposite inequality for supplied trees: a tree cannot be priced above its unbundled replication because holders could sell the tree and buy the replication without tightening constraints. But the reverse arbitrage is constrained because it would require issuing liabilities to assemble the replicating portfolio.

7.3 Concavity in factor loadings (Proposition 5)

Define the tree's gross return:

$$R_{t+1}(\delta \mid s^{t+1}) \equiv \frac{\delta_{t+1}(s^{t+1}) + P_{t+1}(\delta \mid s^{t+1})}{P_t(\delta \mid s^t)}.$$

Project it on K factors with mean zero:

$$R_{t+1}(\delta \mid s^{t+1}) = \mathbb{E}_t[R_{t+1}(\delta \mid s^{t+1})] + \sum_{k=1}^{K} \beta_k(\delta \mid s^t)\, F_{k,t+1}(s^{t+1}) + \varepsilon_{t+1}(\delta \mid s^{t+1}), \tag{28}$$

$$0 = \mathbb{E}_t[F_{k,t+1}] = \mathbb{E}_t[\varepsilon_{t+1}] = \mathbb{E}_t[F_{k,t+1}\varepsilon_{t+1}]. \tag{29}$$

Let agent i's stochastic discount factor be

$$M^i_{t+1}(s^{t+1}) \equiv \frac{Q^i_{t+1}(s^{t+1})}{\pi_{t+1}(s_{t+1} \mid s^t)}.$$

Proposition 5 (Proposition 5: concave beta pricing)**.** *If the residual is orthogonal to all agents' SDFs,*

$$\mathbb{E}_t\big[M^i_{t+1}\, \varepsilon_{t+1}(\delta)\big] = 0 \quad \textit{for all } \delta \textit{ and all } i, \tag{30}$$

then the expected return $\mathbb{E}_t[R_{t+1}(\delta)]$ *is a* ***concave function*** *of the factor exposures* $\{\beta_k(\delta)\}_{k=1}^K$.

Intuition. The equilibrium price is an upper envelope of linear functionals (26), hence *convex* in payoffs. Since expected return is inversely related to price, it becomes *concave* in exposures to risks priced by different agents.

8 Optimal payoff sets vs. optimal tree sets (Section II.F)

One-period-ahead payoffs depend on future prices, so segmentation in payoff space is endogenous. The paper shows an equivalent representation directly over *dividend streams*.

Let J denote a *sequence of agent types* along histories (a mapping from nodes to types). Define recursively a sequence-specific valuation kernel:

$$q_{t+1}(s^{t+1} \mid J) = q_t(s^t \mid J) \times Q_{j_t(s^t),t+1}(s_{t+1} \mid s^t), \qquad q_0(s_0 \mid J) = 1.$$

Then:

$$P_t(\delta \mid s^t) = \max_J \sum_{(u,s^u)\succ(t,s^t)} \frac{q_u(s^u \mid J)}{q_t(s^t \mid J)}\, \delta_u(s^u). \tag{31}$$

Interpretation. Tree pricing includes an endogenous **resale option**: a tree can be bought today by one type and later sold to another type with higher valuation for the remaining cash flows. This is analogous to heterogeneous-beliefs resale-option pricing, but here heterogeneity is generated by incentive constraints.

Implication for segmentation. The "optimal tree set" for an agent is not necessarily convex: it is a union of optimal sets over all sequences J starting with that agent.

9 Conclusion

9.1 Summary

This paper concludes that **imperfect pledgeability of asset payoffs generates endogenous incompleteness**, even when the set of traded claims is rich (including one-step Arrow securities). Because agents cannot fully pledge the payoffs of the assets used as collateral, they face **incentive constraints** that restrict state-contingent borrowing. These constraints prevent full risk sharing, so agents end up with **different effective intertemporal marginal rates of substitution** and therefore **different valuations of the same assets**.

Three broad equilibrium implications follow:

(i) **Endogenous segmentation in asset holdings.** Since agents value payoffs differently, they optimally hold different assets: each asset is held by the agents who value its payoff profile the most. Segmentation arises *endogenously* from incentive constraints, not from exogenous market incompleteness. The paper characterizes this segmentation using a geometry/assignment representation (optimal transport / power-diagram logic).

(ii) **A one-directional basis.** The equilibrium price of a security can be *lower* than the price of a replicating portfolio made of *long positions*, because replication typically requires rearranging payoffs across states in a way that would demand state-contingent borrowing (or otherwise violates pledgeability constraints). Importantly, the paper shows the basis has a sign: the price of the bundled security is not above the long-only replicating portfolio price. This creates systematic "cheapness" of some securities relative to replication, rather than symmetric arbitrage deviations.

(iii) **Concavity of expected returns in factor loadings.** Because equilibrium prices are formed as an *upper envelope* of heterogeneous private valuations, expected returns are **concave** (not linear) in factor exposures. Hence, the standard linear beta-pricing intuition can fail in segmented markets created by incentive constraints.

9.2 The mechanism in one continuous chain

The core intuition is that **pledgeability splits payoffs into two valuation regimes** and therefore creates heterogeneous pricing kernels.

Step 1: Pledgeability creates an incentive constraint. If only a fraction of asset payoffs can be seized by creditors, then an agent cannot promise state-contingent repayments exceeding the *pledgeable* part of her portfolio payoff. This limits the extent to which she can use Arrow trading to transfer resources across states.

Step 2: Limited state-contingent borrowing implies incomplete risk sharing. Even if Arrow securities exist, the incentive constraint can prevent agents from choosing the Arrow positions that would implement full insurance. Therefore, agents retain heterogeneous consumption exposures and thus heterogeneous marginal utilities across states.

Step 3: Heterogeneous marginal utilities imply heterogeneous private valuations. A convenient representation is that each agent i values an asset using a **private** state-price vector Q^i: the pledgeable part is valued at market Arrow prices, while the non-pledgeable part is valued at the agent's own IMRS. This heterogeneity means that two agents can rationally disagree on the "right" price of the same payoff vector.

Step 4: Segmentation becomes an assignment problem in payoff space. Given heterogeneous Q^i, each payoff vector is allocated to the agent who values it most. Hence assets are held by different clienteles, and these clienteles are shaped by the *direction* of payoffs across states (e.g., "bad-state insurance" versus "good-state exposure") rather than only by payoff scale.

Step 5: Why the basis has a sign. Replication of a payoff bundle using long-only positions effectively *unbundles* the payoff into components that different agents would like to hold. That unbundling can raise value because it allows the economy to allocate each component to the agent with the highest private valuation. However, implementing that unbundling through market trades generally requires state-contingent transfers that are constrained by pledgeability. As a result, the market cannot fully arbitrage away the cheapness of bundled payoffs, leading to a basis that is systematically "in one direction."

Step 6: Why beta pricing becomes concave. Since equilibrium price is the maximum across heterogeneous linear valuation functionals, the price is convex in payoff exposures; expected return, being inversely related to price, becomes concave in factor loadings. Economically, different agents price different parts of risk, and bundling exposures does not aggregate linearly into expected returns.

9.3 Broader implications and extensions emphasized by the paper

While the model is a pure exchange economy, the paper highlights an important direction: **introducing production and investment**. In that extension, production technologies play a *dual role*:

(i) they raise output (technological efficiency), and

(ii) they serve as collateral (collateral efficiency), improving risk sharing and consumption smoothing.

A key possibility is a **trade-off between technological and collateral efficiency**: a technology that boosts production strongly might be poor collateral, and vice versa. This trade-off can break standard separation results and may have implications for: (i) international trade patterns and (ii) the joint evolution of technological and financial development.

9.4 Takeaway

When asset payoffs are imperfectly pledgeable, incentive constraints prevent full risk sharing even with rich securities, generating heterogeneous private state prices that produce endogenous segmentation, a one-direction basis relative to long-only replication, and concave (nonlinear) expected-return relations in factor loadings.